Introduction to Supervised Machine Learning

QLS–MiCM Workshop

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Section: Introduction

Workshop Overview

1. Introduction & Motivation.
2. Linear Regression.
3. Regularization.
4. Feature Engineering & Nonlinearity.
5. MLPs for Tabular Data.
6. Training Mechanics & Evaluation.
7. Mini Hackathon

Why Machine Learning in Medicine?

- Large amounts of biomedical data:
  - Clinical measures, omics, imaging, EHRs.
- ML can:
  - **Predict** outcomes (e.g., risk, response).
  - **Identify** patterns and biomarkers.
  - **Support** clinical decisions.

Exercise 00: Intro & Data Exploration

No modelling yet! Let's explore the data.

What You Need to Know

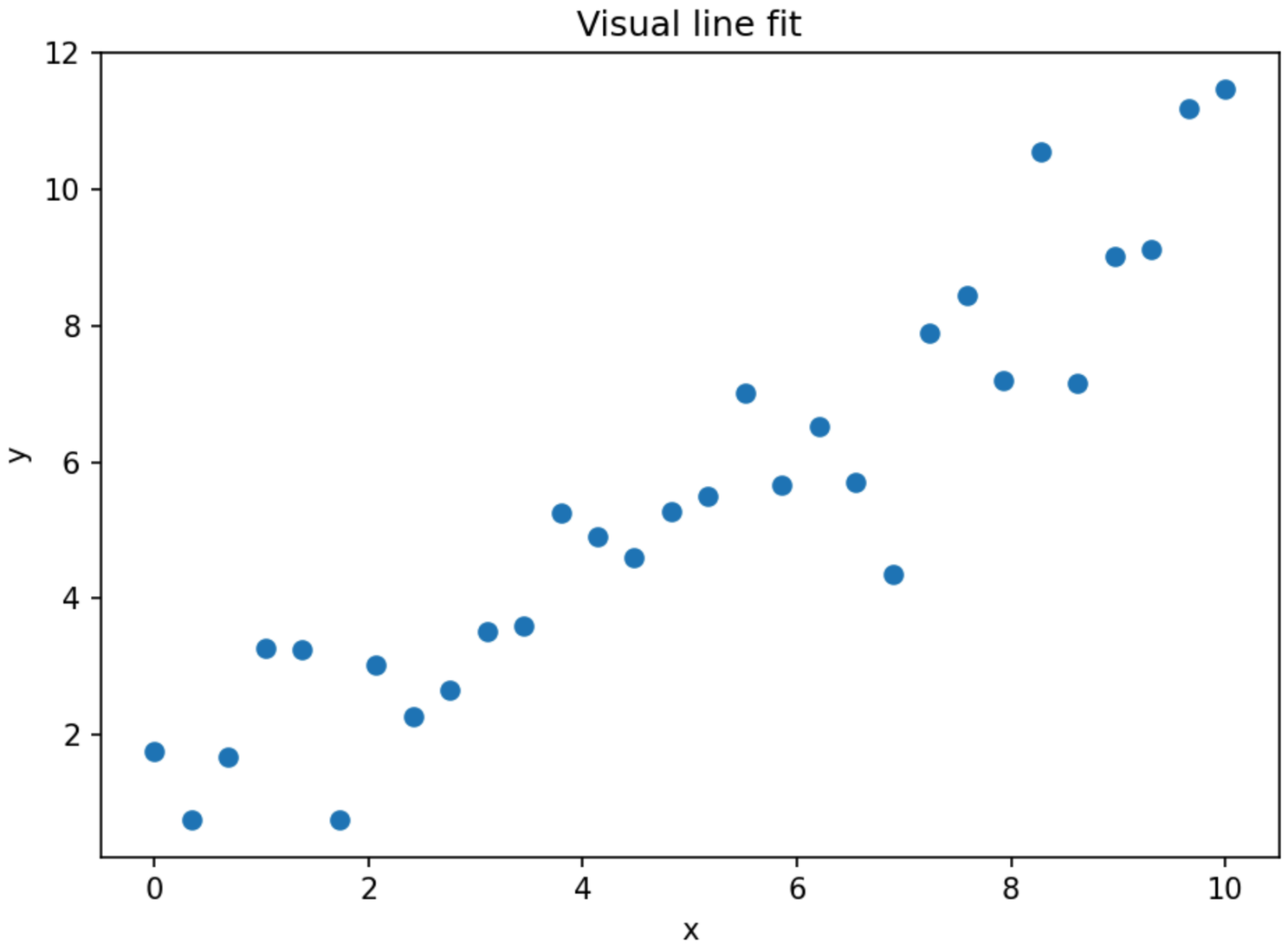
Supervised learning :

- We have **features (X)** and a **target (y)**
- Our goal: learn a function of X that predicts y. `f(X)=y`
- Everything we do today follows this structure.

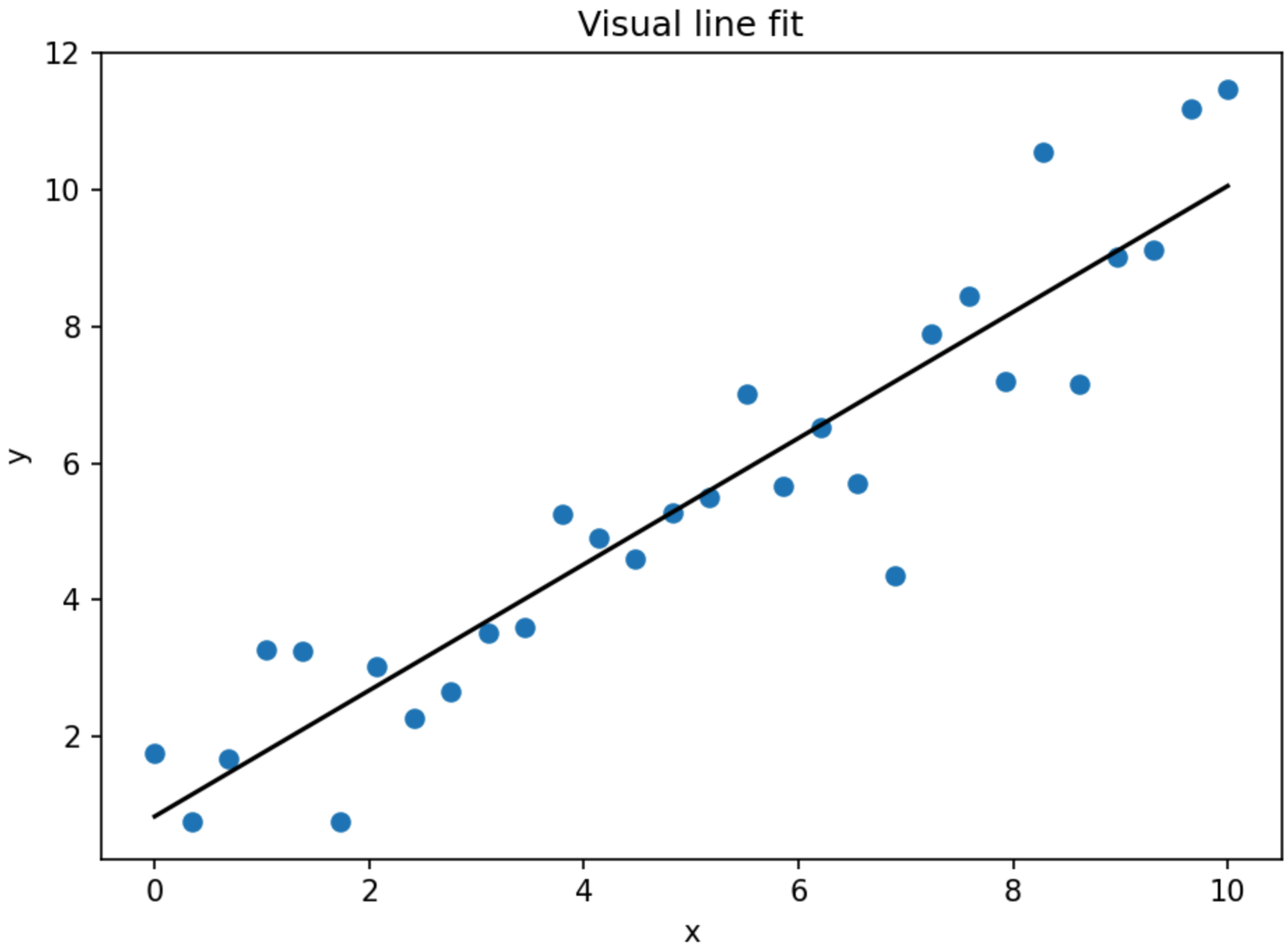
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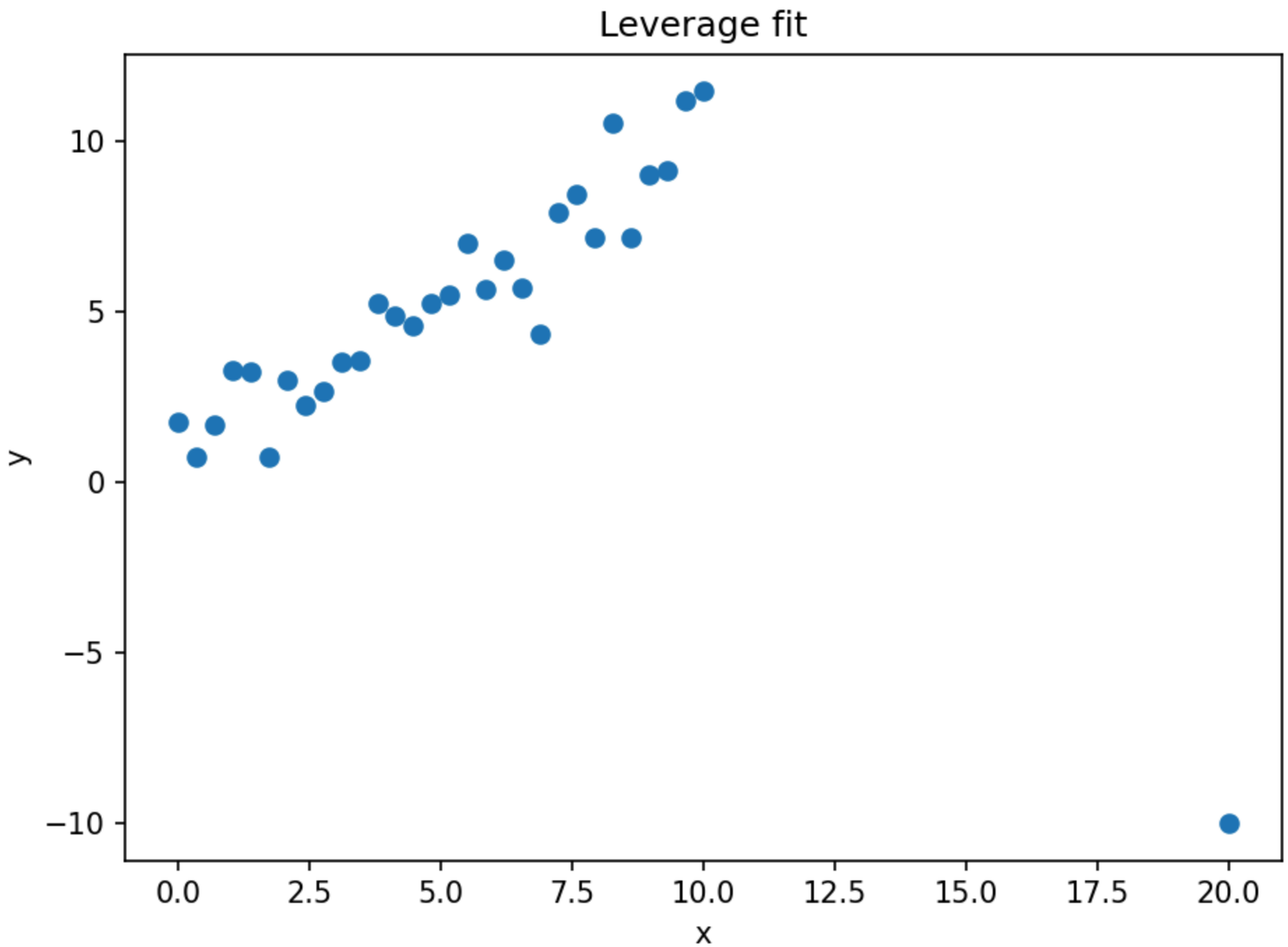
Example 1: A Simple Linear Relationship



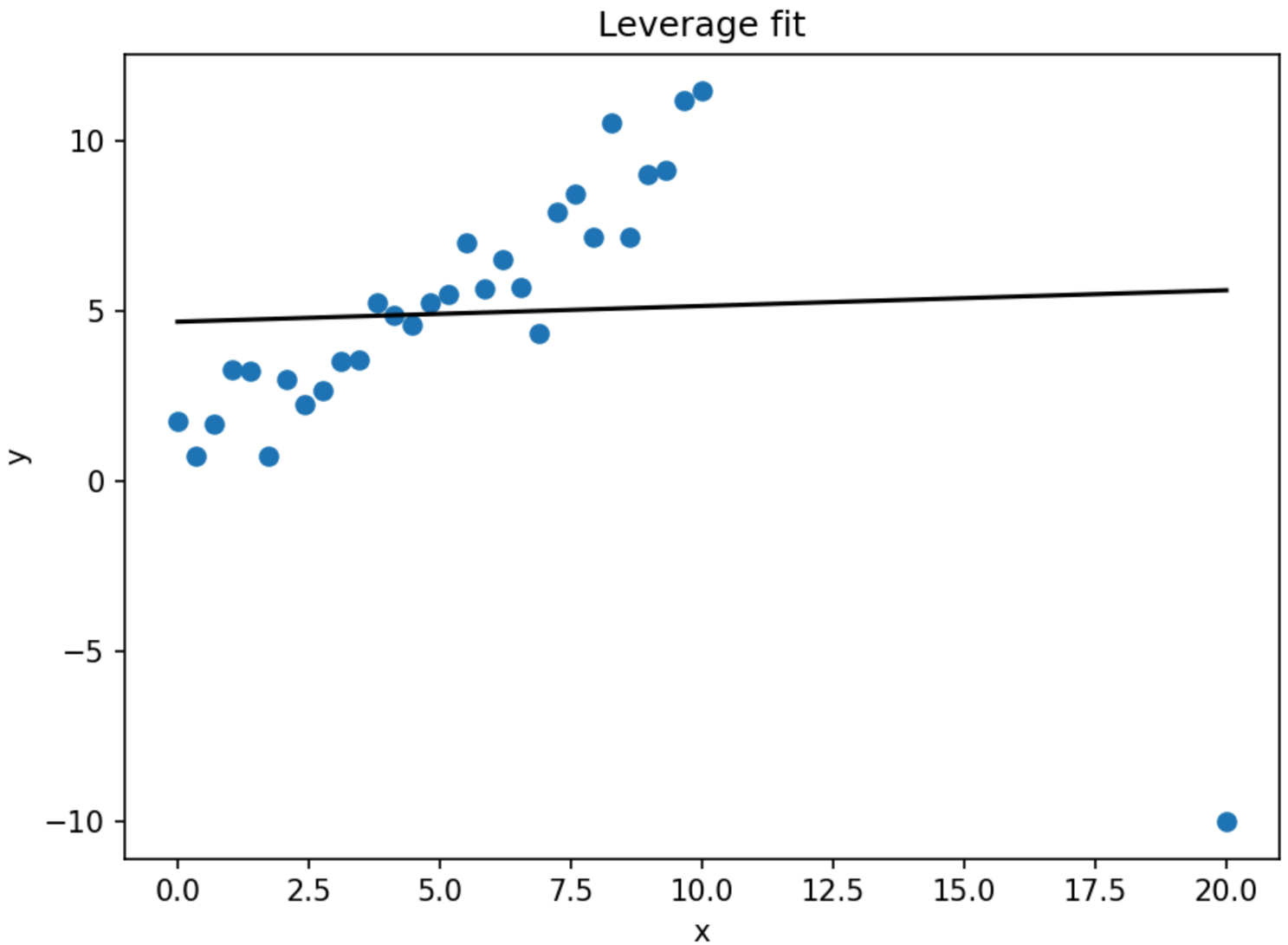
Example 2: Linear Fit Through the Data



Example 3: Adding a High-Leverage Point



Example 4: How the Line Changes



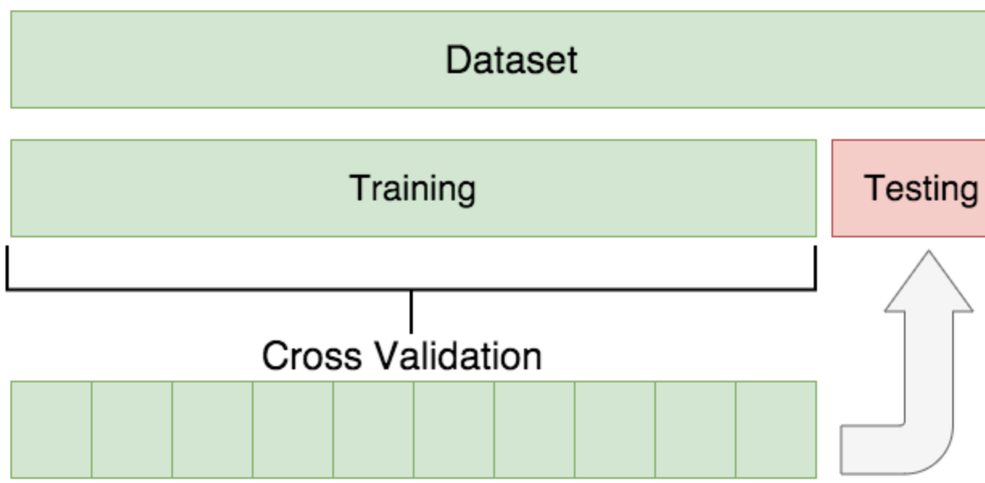
The Workflow We'll Use All Day

1. Load dataset.
2. Inspect structure (shape, head).
3. Explore variables (summary stats).
4. Visualize relationships in the data.
5. Train/test split.

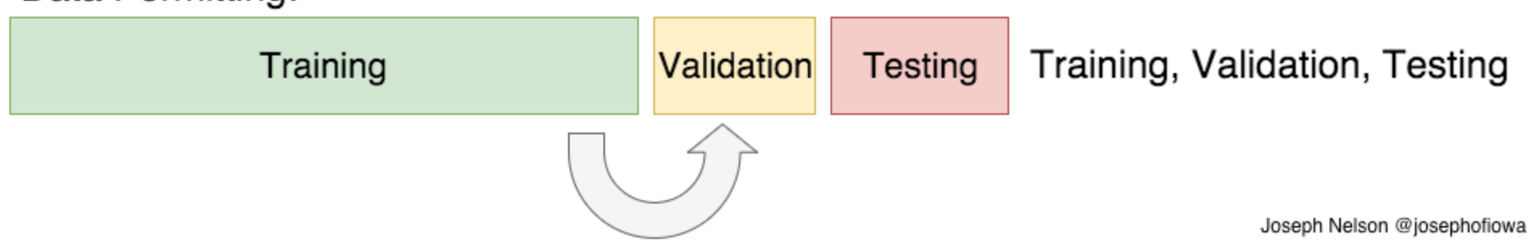
You'll repeat this in every later exercise to understand the data.

Note on Train-Test Split

Important Concept in Supervised Learning:



Data Permitting:



Joseph Nelson @josephofiowa

Your First Dataset

Open the notebook:

`00_intro_and_data_exploration.ipynb`

You'll do:

- `df.head()`.
- `df.shape`.
- `df.describe()`.
- Plot histograms.
- Plot correlations.
- Identify potential predictors

Nothing model-related yet. Just understanding the data.

Why Explore?

Before training any model, we will answer:

- What variables are there?
- Which features look related to the target?

Good exploration = better modeling later.